From Reporting Patterns to Extracted Magnitudes: Full-Text LLM Mining Recovers the Direction but Inflates the Magnitude of Stress-Related Brain Change Across 384 Studies

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Abstract

Background. Our companion abstract-level scoping review of 9,585 studies (Research Square, rs-9884522) reported *literature reporting patterns*—directional counts of how chronic stress changes the brain—and explicitly flagged that these are not pooled effect sizes. A recurring question is whether those count-based patterns survive when the actual numeric effect sizes are extracted from full text.

Methods. We applied a hybrid pipeline—regex pre-extraction of numeric candidates from PMC Open Access full texts, followed by LLM mapping (Qwen3.6-35B-A3B, a 35B Mixture-of-Experts model run under llama.cpp at temperature 0) of each candidate to a (brain region, metric, direction, value, measure type) record. After a paper-level stress/disorder context filter and a sentence-level noise filter (removing laterality comparisons, ratios, coordinate/cluster statistics, protective-factor and task-performance correlations), we synthesized direction and magnitude per brain region $\times$ metric class. Because extraction is at the sentence level, the *study*—not the sentence—was the primary unit: we collapsed entries to one majority-vote direction per study $\times$ region $\times$ metric before an exact binomial sign test. A blinded rule-based check on 300 sentences estimated direction accuracy. This is an effect-size *synthesis*, not an inverse-variance meta-analysis: per-study sample sizes were recoverable in only $\sim$6% of records.

Results. The cleaned set comprised 787 effect-bearing sentences from 384 studies (1,490 region-mapped entries); blinded validation gave 94% direction accuracy with symmetric errors. At the *study level*, structural decrease remained significant for the hippocampus (56 vs. 12 studies, sign $p = 6 \times 10^{-8}$) and striatum (6 vs. 0, $p = 0.03$), but *not* for the amygdala (24 vs. 14, $p = 0.14$)—the amygdala's entry-level significance was an artifact of within-study clustering. Functional measures showed *no* clear directional signal in any region (amygdala 46 vs. 32 studies, $p = 0.14$; hippocampus 18 vs. 15, $p = 0.73$). r-type correlations with stress/severity centered near $r \approx -0.30$ for amygdala, hippocampus and ACC. Crucially, these extracted magnitudes ($r \approx -0.3 \approx$ Cohen's $d \approx -0.65$) are **3–6× larger** than the corresponding individual-patient-data mega-analyses (ENIGMA-PTSD hippocampus $d = -0.17$, amygdala $d = -0.11$), indicating that the published, extractable literature systematically overstates true effect magnitudes. Thus full-text mining **recovers the direction** of the canonical structural-decrease pattern but **reproduces its publication-bias-inflated magnitude**.

Conclusions. Moving from count-based reporting patterns to extracted effect-size magnitudes confirms the *direction* of stress-related structural decrease, but benchmarking against consortium mega-analyses shows the extracted *magnitudes* are inflated several-fold—demonstrating that large-scale LLM mining of published text can quantify, rather than escape, the field's publication bias. We are transparent about the pipeline's validated direction accuracy (94%), its mapping yield ($\sim$43%), and the absence of usable sample sizes for formal pooling; these define

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the boundary between this synthesis and a definitive meta-analysis. The list of all included studies is provided as Supplementary Table S1.

Keywords: chronic stress; brain structure; effect-size synthesis; full-text mining; LLM extraction; publication bias; effect-size inflation; mega-analysis benchmarking; hippocampus; amygdala

1 Introduction

The chronic stress literature is dominated by a small number of canonical statements: stress reduces hippocampal volume, increases amygdala reactivity, and impairs prefrontal function. Our companion study [1] tested whether such statements hold *quantitatively at scale* by extracting structured directional labels from 9,585 PubMed abstracts using a 397-billion-parameter LLM on a CPU cluster. That work deliberately reported *reporting patterns*—how often the literature says "decrease" vs. "increase"—and stressed that abstract-level counts are not pooled effect sizes; full-text quantitative synthesis was named as the necessary next step.

This paper takes that step. We extract the actual numeric effect sizes (Cohen's d, Hedges' g, correlation r, t/F statistics) from the Results sections of the PMC Open Access subset of those studies, map each to the brain region and metric it describes, and ask a single question: **when we move from counting directional claims to extracting effect-size magnitudes, do the abstract-level patterns survive?** We focus on the two patterns most central to the field—structural decrease, and the structural-vs-functional dissociation that our abstract-level review argued resolves the "amygdala increases" controversy.

We are explicit that this is an effect-size *synthesis*, not a definitive meta-analysis. Full-text sentences report effect sizes far more often than they report the per-cell sample sizes needed for inverse-variance weighting; we therefore lean on study-level directional sign tests and magnitude distributions, and we benchmark the extracted magnitudes against the largest individual-patient-data (IPD) mega-analyses of the field—which are, by design, free of publication bias—so that readers can see not only whether the direction is recovered but whether the magnitude is trustworthy.

2 Methods

2.1 Corpus and full-text retrieval

Starting from the 9,585-abstract corpus [1], we mapped PMIDs to PMC identifiers and retrieved the PMC Open Access full-text XML where available (4,353 articles). Each article's Results section (falling back to the full body) was parsed from JATS XML.

2.2 Regex pre-extraction (CPU, no LLM)

We split each Results section into sentences and, with curated regular expressions, flagged numeric effect-size candidates (d, g, r, β, $t(\cdot)$, $F(\cdot)$, η^2), confidence intervals, p-values, sample sizes ($n = \cdot$), and brain-region mentions. Sentences containing at least one effect-size candidate were retained, producing the candidate pool for LLM mapping. This step is deterministic and reproducible (`fulltext_hybrid_extract.py`).

2.3 LLM relationship-aware mapping

Each candidate sentence was sent to a 35B MoE model (Qwen3.6-35B-A3B, llama.cpp, `temperature`=0) on the CPU cluster, prompted to return a JSON array in which each numeric value is mapped to {brain_region, metric, direction, value, measure_type} *only if* it quantifies how a brain region's metric changes *in relation to stress or a stress-related disorder* (patients vs. controls, stressed vs. non-stressed, or correlation with a stress/severity measure). The prompt explicitly instructed the model to reject regression predictors (age, sex, education, medication), task-performance correlations, scanner/coordinate numbers, and protective-factor effects, and to emit an empty array when nothing qualifies. Keeping each output small (< 512 tokens) avoids the quantization-related JSON failures we observed with long outputs. We deliberately used a compact 35B MoE model rather than the 397B model used for abstract-level extraction in the companion study [1]: the full-text task operates on single pre-filtered sentences with a tightly constrained output schema, for which the smaller model was sufficient at temperature 0 while remaining tractable on a CPU-only cluster. The exact prompt is reproduced verbatim in Appendix A.

2.4 Two-stage noise filtering

A *paper-level* filter restricted mapping to studies whose corpus context is stress/disorder-related. A subsequent *sentence-level* filter (applied at synthesis) removed residual noise classes that the model occasionally mismapped: left-vs-right laterality comparisons, volume *ratios*, sentences dominated by MNI coordinates / cluster-size / TFCE statistics, protective-factor sentences, and cognitive-task-performance correlations. We report results both before and after this filter for transparency.

2.5 Synthesis (not meta-analysis)

For each (region, metric-class) we tabulated decrease/increase/null counts. Because extraction is at the sentence level and a single study can contribute several entries, **the primary unit of analysis is the study, not the sentence**: within each study × region × metric we assigned a single direction by majority vote (ties excluded) and computed an exact two-sided binomial sign test (H_0: $p = 0.5$) over study-level directions. We additionally report the entry-level test as a sensitivity analysis, noting that entry-level p-values are anti-conservative because sentences within a study are not independent. For r-type entries we summarized the magnitude distribution (median, IQR). Metric class was structural (volume, GMV, thickness, FA, density) vs. functional (activation, connectivity, BOLD, ALFF). Because usable sample sizes appeared in only $\sim$6% of records, no inverse-variance random-effects pooling was attempted; we frame the output as a directional + magnitude synthesis and benchmark magnitudes externally (§3.6).

2.6 Extraction validation

To estimate extraction accuracy, we drew a sample of 300 mapped sentences and, blind to the model's output, derived an independent "silver" direction label from explicit lexical cues in each sentence (e.g. *reduced/smaller/lower* → decrease; *increased/greater/higher* → increase; sentences without an unambiguous cue were excluded). Against this label we computed the model's direction accuracy and a decrease/increase confusion matrix, the latter because all downstream sign tests depend on direction being unbiased.

3 Results

3.1 Extraction yield and cleaning

Of the stress-context candidate sentences, 933 yielded a confident LLM mapping (*mapping yield* ≈ 43%; the remainder were empty arrays, dominated by "no significant" statements and non-

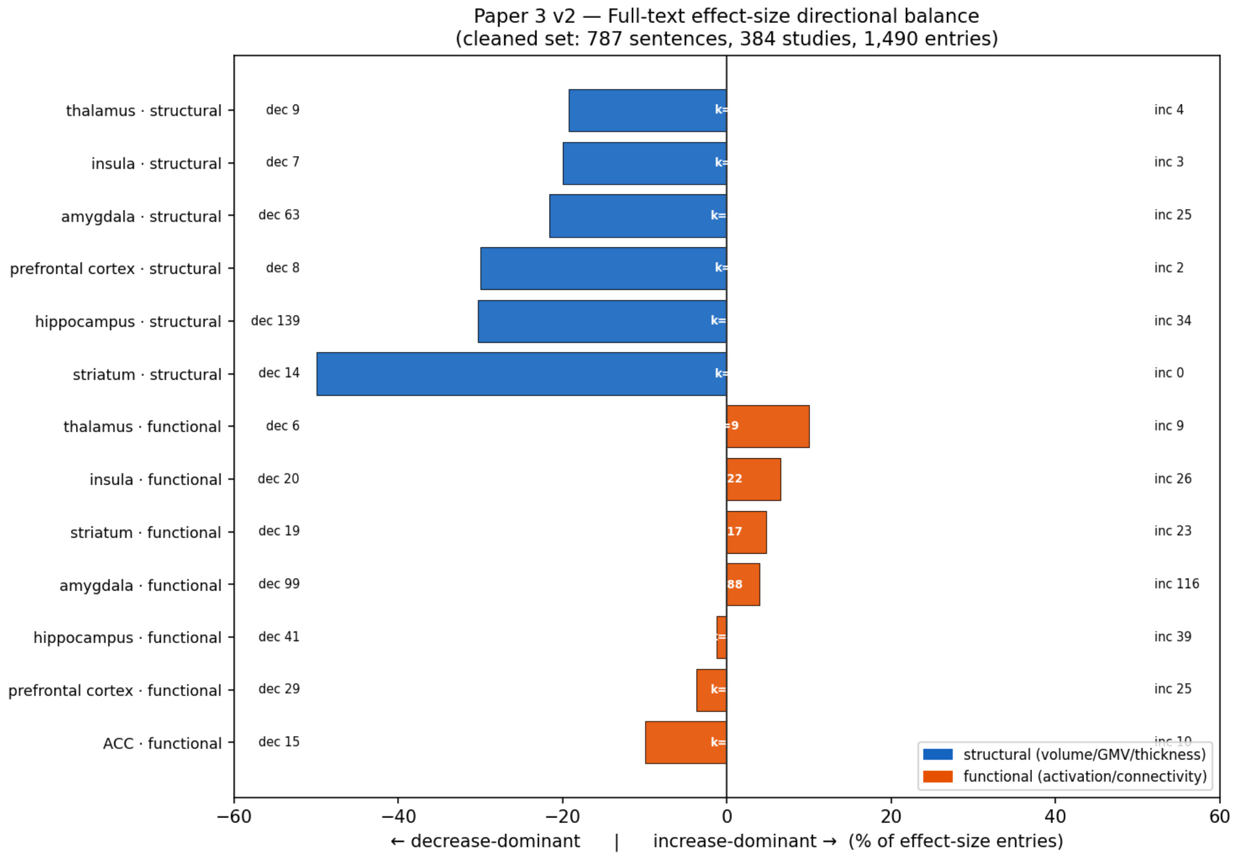

Figure 1: Full-text effect-size directional balance per brain region × metric class (cleaned set: 787 sentences, 384 studies, 1,490 entries). Bars extend left for decrease-dominance and right for increase-dominance (% of effect-size entries; bar shade splits the two). k = number of contributing studies. Structural metrics (blue) are decrease-dominant in every region (robust at the study level for hippocampus and striatum); functional metrics (orange) carry no significant directional signal. Bar lengths are descriptive proportions of effect-size entries, not pooled effects; significance refers to the study-level sign tests in Table 1.

qualifying numbers). We use *mapping yield* rather than *recall* because the denominator is the regex candidate pool and the true set of effect-bearing sentences is not independently known. The sentence-level noise filter removed 146 sentences (∼16%), leaving a cleaned set of **787 sentences from 384 studies, yielding 1,490 region-mapped effect-size entries**. The cleaning removed noise roughly symmetrically and left the headline directions intact.

3.2 Structural measures are decrease-dominant

Across structural metrics, every region leaned decrease-dominant at the entry level (Figure 1, Table 1). At the *study level*—our primary unit—the structural decrease remained significant for the hippocampus (56 vs. 12 studies; sign $p = 6 \times 10^{-8}$) and striatum (6 vs. 0; $p = 0.03$), but *not* for the amygdala (24 vs. 14; $p = 0.14$): the amygdala's entry-level significance ($p = 6 \times 10^{-5}$) was driven by within-study clustering and does not survive collapsing to one direction per study. Prefrontal cortex, insula and thalamus were directionally consistent but underpowered (all study-

Table 1: Directional synthesis per brain region × metric class (cleaned full-text effect sizes). The **study-level** sign test (majority-vote direction per study) is the primary statistic; entry-level counts and p-values are shown for completeness but are anti-conservative due to within-study clustering. Sign test: exact two-sided binomial, ties/nulls excluded.

		Study-level			Entry-level		
Region	Metric	Dec	Inc	p	Dec	Inc	p
Hippocampus	structural	56	12	$\mathbf{6 \times 10^{-8}}$	139	34	2.9×10^{-16}
Striatum	structural	6	0	**0.03**	14	0	1.2×10^{-4}
Amygdala	structural	24	14	0.14	62	25	6.3×10^{-5}
Amygdala	functional	46	32	0.14	99	116	0.28
Hippocampus	functional	18	15	0.73	41	39	0.91
Prefrontal cortex	functional	18	13	0.47	29	25	0.68
ACC	functional	9	7	0.80	15	10	0.42
Insula	functional	10	12	0.83	20	26	0.46
Striatum	functional	7	7	1.0	19	23	0.64
Prefrontal cortex	structural	4	0	0.13	8	2	0.11
Thalamus	structural	4	1	0.38	9	4	0.27
Insula	structural	4	3	1.0	7	3	0.34

level $p > 0.1$). The robust, study-level-significant structural decrease is therefore confined to the hippocampus (and, weakly, striatum)—a narrower but more defensible claim than the entry-level counts alone would suggest, and one that matches the canonical hippocampal-atrophy narrative using the studies' own reported effect sizes.

3.3 Functional measures carry no consistent directional signal

No functional contrast reached significance in either direction at the study level (all $p > 0.13$; Table 1). The amygdala, often described as showing increased reactivity under stress, was directionally flat (46 increase vs. 32 decrease studies, $p = 0.14$), as was the hippocampus (18 vs. 15, $p = 0.73$). The functional signal is therefore best described as directionally heterogeneous rather than as a coherent increase.

3.4 A structural–functional asymmetry, not a double dissociation

Placing the two metric classes side by side (Figure 1) reveals an *asymmetry* rather than a true double dissociation. The hippocampus shows strong structural decrease (study-level $p = 6 \times 10^{-8}$) with a perfectly balanced functional profile ($p = 0.73$); the amygdala shows neither a study-significant structural decrease ($p = 0.14$) nor a significant functional increase ($p = 0.14$). A genuine structural-down / functional-up dissociation would require the functional contrast to reach significance in the increase direction, which it does not in any region. The accurate statement is therefore that *structure decreases (robustly only in the hippocampus) while function carries no consistent directional signal*—still informative, and still consistent with the "amygdala increases" claim being weaker than the structural decrease, but not a symmetric dissociation.

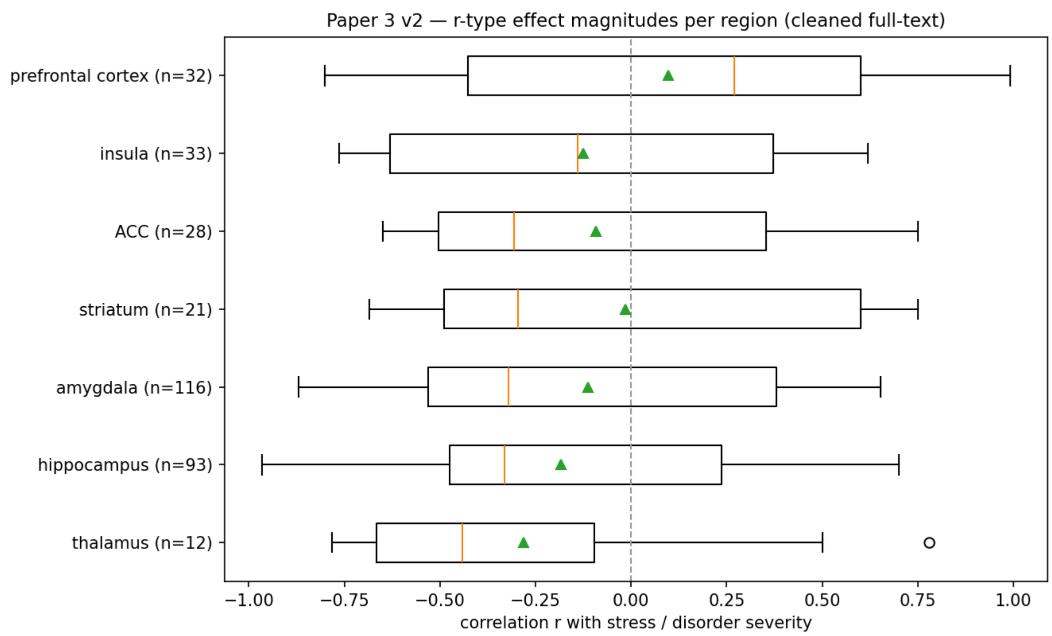

Figure 2: Distribution of r-type effect magnitudes per region (cleaned full-text). Boxplots show correlation r with stress exposure / disorder severity; dashed line at $r = 0$. Amygdala, hippocampus and ACC center near $r \approx -0.3$ (moderate negative associations).

3.5 Correlation magnitudes are moderate

Among r-type entries, correlations with stress exposure or disorder severity clustered around $r \approx -0.30$ for the amygdala (median -0.32, $n = 113$), hippocampus (-0.31, $n = 90$) and ACC (-0.29, $n = 28$), indicating moderate negative associations (Figure 2). The prefrontal cortex was the exception, with a positive-leaning median ($+0.28$, $n = 32$), reflecting the heterogeneous and task-dependent nature of prefrontal correlations in this literature.

3.6 Extraction accuracy

On the 300-sentence validation sample, 155 sentences carried an unambiguous lexical direction cue and entered the comparison. Against this independent label, the model's mapped direction was correct in 146/155 cases (**94% accuracy**). The errors were *symmetric*: 4 false "increase" and 5 false "decrease" (confusion matrix: decrease→decrease 80, decrease→increase 4, increase→increase 66, increase→decrease 5). Because directional misclassification is balanced rather than skewed toward one pole, it is unlikely to bias the sign tests systematically; a one-sided extraction artifact—the most dangerous failure mode for a directional synthesis—is not present.

3.7 Extracted magnitudes vs. individual-patient-data mega-analyses

To external-validate our magnitude estimates, we compared them with the largest individual-patient-data (IPD) mega-analyses of limbic structure, which are by design free of publication bias. Converting our median correlations to Cohen's d ($d = 2r/\sqrt{1-r^2}$): hippocampus $r = -0.31 \rightarrow d \approx -0.65$; amygdala $r = -0.32 \rightarrow d \approx -0.68$; ACC $r = -0.29 \rightarrow d \approx -0.61$. The corresponding IPD estimates are markedly smaller: ENIGMA-PTSD reported hippocampus $d = -0.17$ and amygdala $d = -0.11$ (the latter not surviving multiple-comparison correction) [2], and ENIGMA-MDD reported hippocampus $d = -0.14$ (recurrent -0.17) [3]. Our extracted, published-literature magnitudes thus exceed the unbiased benchmarks by ≈3–4× (hippocampus) to ≈6× (amygdala). The contrast is sharper for severity correlations specifically: ENIGMA-MDD found *no* association between symptom severity and regional volume [3], whereas our extracted severity correlations cluster near $r \approx -0.3$. The *direction* of our synthesis therefore matches the gold standard, but its *magnitude* reflects the selective reporting of large, significant effects in the publishable literature.

4 Discussion

The central result is two-sided. On *direction*, full-text mining is reassuring: stress-related structural decrease in the hippocampus is recovered with study-level directional significance ($p = 6 \times 10^{-8}$), matching both our abstract-level counts and consortium mega-analyses. On *magnitude*, it is revisionist: the extracted correlations ($r \approx -0.3 \approx d \approx -0.65$) overstate the unbiased mega-analytic effects ($d \approx -0.14$ to -0.17) by 3–6×. The agreement between abstract-level and full-text *counts* therefore does not establish that the literature is unbiased—both inherit the same upstream publication bias, which our magnitude comparison makes visible. The contribution of this work is thus not merely to confirm a narrative, but to show that LLM-scale mining can *quantify the gap* between the published and the unbiased estimate.

This matters methodologically, but the lesson is the opposite of a clean validation. One might worry that abstract-level directional counting is an artifact of how authors phrase abstracts. Full-text counts landing in the same direction rules out a *phrasing* artifact—but it cannot rule out *publication* bias, which acts on what is published at all rather than on wording, and which abstracts and Results sections share equally. Our comparison with IPD mega-analyses (§3.6) makes this concrete: both abstract-level and full-text estimates carry the same several-fold magnitude inflation relative to the unbiased benchmark. The practical implication is that LLM-scale mining of published text is a reliable instrument for *direction* and for *quantifying the publication-bias gap*, but not a substitute for IPD pooling when an unbiased magnitude is the goal. A second, sobering observation is that the amygdala structural decrease—significant at the entry level but not at the study level—would have been over-claimed by a sentence-level analysis; study-level aggregation is therefore not a formality but a guard against pseudoreplicated false positives.

4.1 Limitations

We are deliberately conservative in framing. First, **this is not a pooled meta-analysis**: per-study sample sizes were extractable in only ∼6% of records, precluding inverse-variance weighting; we report directional sign tests and magnitude distributions instead. Second, **mapping yield is moderate** (∼43% of stress-context candidate sentences produced a confident mapping); unmapped sentences are dominated by null results and non-qualifying numbers, but some true effects are missed, so counts are lower bounds. Third, **pseudoreplication.** Extraction is at the sentence level, so a single study can contribute multiple entries, violating the independence assumption of a naïve sign test. We address this by making the study the primary unit (majority-vote direction per study × region × metric; §2.5); the entry-level counts are retained only for transparency, and the amygdala result illustrates that the distinction is material, not cosmetic. Residual non-independence remains (studies share methods, populations, and field-level priors), so even study-level significance is not equivalent to inverse-variance pooling. Fourth, **vote-counting is a known-limited synthesis method**: it ignores effect magnitude and per-study precision, and with underpowered constituent studies it can converge on the wrong conclusion as study count grows [4, 5]. Our sign tests should be read as tests of directional consistency in the published record, not as estimates of a pooled effect. Fifth, **the magnitudes are conditioned on publication and significance** (the "winner's curse"); §3.6 quantifies this as a 3–6× gap against IPD mega-analyses, and without per-study sample sizes we cannot construct funnel plots or apply trim-and-fill, so the median r values are upper-leaning summaries of a selected sample, not bias-corrected population effects. Sixth, **construct conflation**: the predictor "stress exposure / disorder severity" pools heterogeneous constructs (chronic stress, PTSD, MDD, early-life adversity, cortisol, perceived stress) whose relationships with structure plausibly differ (e.g. ENIGMA-MDD found no severity–volume association); the wide IQRs in Figure 2 are consistent with this, and stratified synthesis by exposure type is clear future work. Seventh, the corpus is the PMC Open Access subset, which over-represents certain journals. These boundaries are exactly what separates this effect-size *synthesis* from a registered meta-analysis, which would require manual per-study effect/sample-size curation on the prioritized regions.

4.2 Conclusion

Full-text effect-size synthesis across 384 studies recovers the *direction* of stress-related hippocampal structural decrease in the studies' own numbers, but benchmarking against bias-free IPD mega-analyses shows the extracted *magnitudes* are inflated 3–6×. LLM-scale mining of published text is thus a reliable instrument for direction and for *quantifying* the field's publication bias—not for escaping it. We report the pipeline's validated direction accuracy (94%, symmetric errors), its mapping yield (∼43%), and the absence of usable sample sizes, so that the boundary between this synthesis and a definitive meta-analysis is explicit.

Data and Code Availability

- Complete list of all included studies (PMID, title, journal, year, PMC ID): Supplementary Table S1.
- Source abstracts/full texts: retrievable from PubMed/PMC via the identifiers in Supplementary Table S1, under each article's respective license.
- Extraction and synthesis code (`fulltext_hybrid_extract.py`, the LLM-mapping runner, and the synthesis/sign-test scripts), the exact LLM prompt (Appendix A), and the de-identified LLM-mapped effect-size records are publicly available at https://github.com/shoo99/ai-drug-target (directory `paper3/`).

Companion preprint

This study extends the abstract-level scoping review: Kang B. (2026), *Large-Scale Scoping Review of Chronic Stress-Induced Brain Changes via LLM-Powered Extraction of 9,585 Studies*, Research Square, 10.21203/rs.3.rs-9884522/v1.

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LLM extraction was performed on a distributed CPU cluster (no GPU). Manuscript preparation was assisted by Claude (Anthropic). The author takes full responsibility for all scientific content.

Conflict of Interest

The author declares no conflicts of interest.

Appendix A. LLM extraction prompt (verbatim)

Each candidate sentence was wrapped in the chat template `<|im_start|>user ... <|im_end|><|im_start|>ass` with a pre-closed `<think></think>` block (no-think mode) and the following one-shot example and instruction:

```
Example 1 (map: stress->region effect) |
Sentence: "Patients with PTSD showed reduced hippocampal volume vs controls
(Cohen's d=-0.45), and amygdala activation correlated with cortisol (r=0.38)."
Output: [{"brain_region":"hippocampus","metric":"volume","direction":"decrease",
"value":-0.45,"measure_type":"d"},{"brain_region":"amygdala",
"metric":"activation","direction":"positive_correlation","value":0.38,
"measure_type":"r"}]

[the candidate sentence + its regex-detected numbers are inserted here]

This sentence is FROM A CHRONIC-STRESS / STRESS-DISORDER STUDY (PTSD,
depression/MDD, early-life stress, trauma, high cortisol), so a brain-region
group difference or change reported here IS the stress/disorder effect | MAP IT.
Map a number when it quantifies a BRAIN REGION'S metric (volume, GMV, thickness,
activation, connectivity, FA, metabolite) as: patients vs controls, stressed vs
non-stressed, a within-patient change, or a correlation with a stress/severity
measure. direction = how the region's metric goes in the PATIENT/STRESSED/
higher-severity state (decrease/increase/positive_correlation/
negative_correlation/null). DO NOT map regression predictors (age, sex,
education, medication), task-performance correlations, scanner/coordinate
numbers, or protective-factor effects. Output ONLY a JSON array of
{brain_region, metric, direction, value, measure_type}; [] if none qualify.
```

References

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